International Journal of Medicinal Mushrooms, 17(5): 443–452 (2015)

Chemical Compositions and Macrophage Activation of Polysaccharides from Leon's Mane Culinary-Medicinal Mushroom *Hericium erinaceus* (Higher Basidiomycetes) in Different Maturation Stages

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ABSTRACT: We studied the effect of the maturation stage on the chemical compositions and macrophage activation activity of polysaccharides from the culinary-medicinal mushroom *Hericium erinaceus*. Results showed that total polysaccharides increased, whereas protein content decreased with the maturation stage development of fruiting body. Nine polysaccharide fractions, 3 from each of the maturity stages IV (small fungal spine stage), V (mid-fungal spine stage) and VI (mature), were prepared using the gradient ethanol precipitation method. The polysaccharide fraction HP4A isolated from the maturating-stage (stage IV) fruiting body had a significant difference from the fractions HP5A (stage V) and HP6A (stage VI) in the molecular weight distribution and monosaccharide compositions. Immunostimulating tests revealed that the polysaccharide fraction HP6 isolated from the mature stage (stage VI) fruiting body presented higher macrophage activation activity. Our findings provided important information for the harvest and use of *H. erinaceus* with higher qualities and functional benefits.

KEY WORDS: medicinal mushrooms, *Hericium erinaceu*s, fruit body, maturation stage, chemical composition, macrophage activation

ABBREVIATIONS: ATCC, American Type Culture Collection; **HAPEC,** high-performance anion-exchange chromatography; **HPLC,** high-pressure liquid chromatography; **NO,** nitric oxide; **PBS,** phosphate buffer solution; **PDA,** potato dextrose agar; **TFA,** trifluoroacetic acid; **TNF-α,** tumor necrosis factor-alpha

I. INTRODUCTION

Leon's mane culinary-medicinal mushroom, *Hericium erinaceus* (Bull.:Fr.)Pers. (Hericiaceaae, higher Basidiomycetes),[1,2] has attracted considerable attention in Asia recently owing to its effectiveness in treating gastric and duodenal ulcers, chronic gastritis, and other related diseases of the digestive tract, and its neuroregenerative and neuroprotective protective effects.[3] As a consequence, numerous bioactive compounds with immunomodulatory,[4] antitumor,[5] antioxidant,[6] and antimicrobial[7] properties have been isolated from *H. erinaceus* fruit bodies, including polysaccharides,[8,9] laccase,[10] proteins,[11] and hericenone.[12]

Bioactivity associated with macromolecular compounds derived from mushroom fruit bodies has been shown to be closely related to the physicochemical properties of the latter, such as polysaccharide content, protein content, molecular weight distribution, and chemical composition.[13] Furthermore, these properties are significantly influenced by the stage of fruit body development.[14] For example, the levels of protein and monounsaturated and polyunsaturated fatty acids in fruit bodies of a *Lactarius* sp. increased, whereas the polysaccharide and saturated fatty acid contents decreased, as the fruit body reached maturation stage. A direct correlation between the amounts of bioactive compounds present in immature and mature *Lactarius* sp. fruit bod-

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ies and antimicrobial activity was also recorded.[15] In a separate study, the stage of *Agaricus brasiliensis* fruit body development was linked to the yield, structural characteristics, and pharmacological activity of extracted polysaccharides.[16] Hence, knowledge of the changes in the composition of bioactive components and their associated properties occurring during different stages of fruit body development is essential for achieving the highest market value and maximizing potential therapeutic benefits. Because, in this context, very few studies have been undertaken with *H. erinaceus*, we compared total polysaccharide and protein levels in mushroom fruit bodies at 6 developmental stages; evaluated the effect of fruit body development on the yield, monosaccharide composition, and molecular weight distribution of extracted polysaccharide; and investigated possible links between polysaccharide composition and the extent of nitric oxide (NO) release from macrophages.

II. MATERIALS AND METHODS

A. Mushroom Material

Hericium erinaceus, strain H0605, was obtained from the Herbarium of the Edible Fungi Culture Collection Center Branch of the China Culture Collection of Agricultural Microorganisms and maintained on potato dextrose agar (PDA) slants at 4°C with periodic transfer.

B. Fruit Body Cultivation and Selection

Hericium erinaceus was cultivated in polypropylene bags filled with solid medium consisting of (%): 30 sawdust, 16 corncobs, 22 cottonseed hulls, 15 wheat bran, 5 corn starch, 10 rice bran, and 2 plasters. After inoculation with fungal mycelium, the bags were kept in the dark at 25°C and 70% relative humidity for 25 days before transfer to a ventilated field at 18°C and 90% relative humidity for 7 days. Fruit bodies were classified and harvested according to the following stages of maturation: stage I (bud-forming stage), stage II (small lump stage), stage III (split stage), stage IV (small fungal spine stage), stage V (mid-fungal spine stage), and stage VI (mature) (Fig. 1).

C. Total Polysaccharide and Protein Content

Dried fruiting bodies (10 g) at various stages of development were extracted with 180 mL of distilled water at 100°C for 2 hours. Extracts (6 from each of 6 maturation stages) were concentrated at threefold decompression, centrifuged (12,000 × *g*, 30 minutes) and absolute ethanol was added to the supernatant fractions to a final concentration of 70%. After standing for 12 hours at 4°C, precipitated material was collected by centrifugation (4,000 × *g*, 20 minutes), washed twice with absolute ethanol, dialyzed (cutoff 3500 Da) against water for 3 days at 4°C, and freeze-dried. Total polysaccharide contents were determined spectrophotometrically at 490 nm using the phenol-sulfuric acid method with glucose as the standard. Protein was determined using the Kjeldahl method.

D. Polysaccharide Fractionation and Yield Determination

The procedure for fractionating polysaccharide from *H. erinaceus* fruit bodies at different maturation stages is shown in Fig. 2. Air-dried fruit bodies (500 g) from stages IV, V, and VI were extracted 3 times with 9 L of distilled water at 100°C for 4 hours. Combined filtrates were concentrated threefold under reduced pressure, absolute EtOH was added to a concentration of 30%, and the precipitated material was collected by centrifugation (4,000 x *g*, 30 minutes). After washing twice with 30% EtOH, precipitates were dissolved in distilled water and freeze-dried. Freeze-dried samples were re-dissolved in distilled water and dialyzed (cutoff 3500 Da) against water for 3 days at 4°C to yield 3 fractions, HP4A, HP5A, and HP6A, corresponding to the Stages IV, V, and VI, respectively. Absolute EtOH was added to each of these fractions to a final concentration of 50%, and the precipitated material (HP4B, HP5B, and HP6B, respectively) collected by centrifugation. Finally, the EtOH concentration

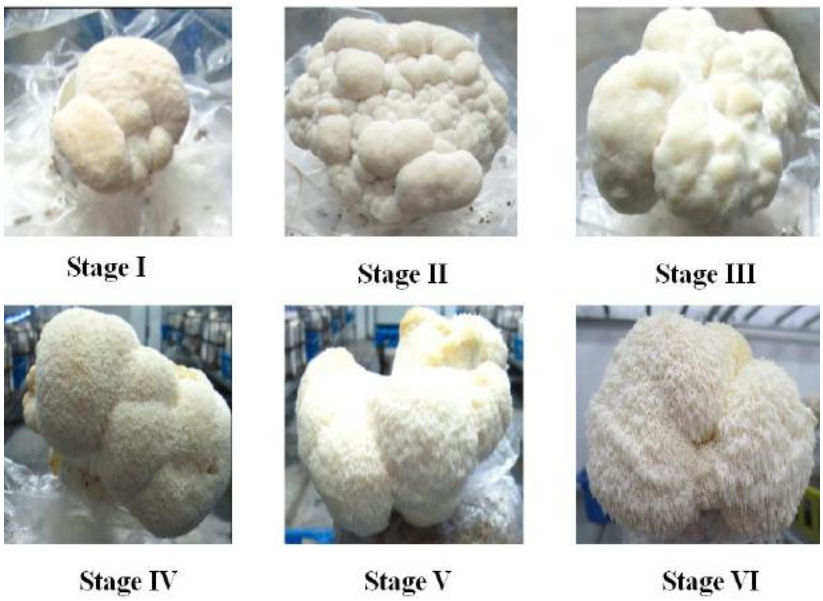

FIG. 1: *Hericium erinaceus* fruit bodies at different stages of maturity: stage I (bud-forming stage), stage II (small lump stage), stage III (split stage), stage IV (small fungal spine stage), stage V (mid-fungal spine stage), stage VI (mature).

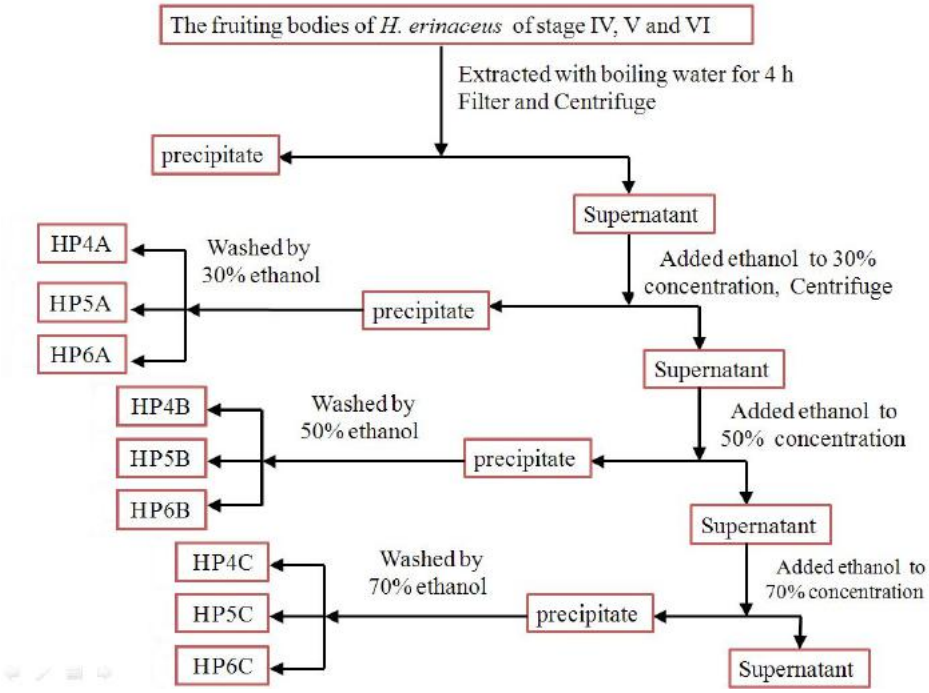

FIG. 2: Summarized fractionation scheme of polysaccharides from *Hericium erinaceus* fruit bodies harvested at different maturation stages**.**

in the supernatants was increased to 70%, and the collected precipitates designated HP4C, HP5C and HP6C, respectively.

Polysaccharide contents in the 9 fractions were determined using the phenol-sulfuric acid method, and yields were calculated as follows:

$$\text{Yield (\%)} = \frac{M_1}{M_2} \times 100,$$

where M1 is the weight of the polysaccharide fraction and M2 is the weight of dried fruit body.

E. Molecular Weight Distribution Among the Polysaccharide Fractions

Molecular weight distribution patterns among the 9 polysaccharide fractions were determined by high-pressure liquid chromatography (HPLC;Waters, Milford, MA, USA) using a Waters 2695 HPLC system fitted with 3 serially linked TSK PWXL 6000, 4000, and 3000 gel filtration columns, a Model 2695 pump, a Waters 2414 RI detector, a Waters 2996 dual wavelength absorbance detector, and an online de-gasser. Columns injected with 20 μL of polysaccharide (2 mg/mL) were eluted with 0.1 M phosphate buffer solution (PBS) and 0.3 M $NaNO_3$ (pH 7.0) at a flow rate of 0.5 mL/min and calibrated using Dextrans T-140, 67, 27, 15, 8, 2.5, 1.2, 0.5 (Sigma-Aldrich, St Louis, MO, USA; corresponding molecular weights, approximately 1.4×10^6 Da, 6.7×10^5 Da, 2.7×10^5 Da, 1.5×10^5 Da, 8×10^4 Da, 2.5×10^4 Da, 1.2×10^4 Da, 5×10^3 Da).

F. Monosaccharide Composition of Polysaccharide Fractions

Monosaccharide components in the 9 polysaccharide fractions and their relative ratios were determined after hydrolysis with 2 M trifluoroacetic acid (TFA) at 110°C for 3 hours using high-performance anion-exchange chromatography (HAPEC) and D-Gal, D-Glc, D-Ara, L-Fuc, L-Rha, D-Man, D-Xyl, D-Fru, Rib, D-GluA, and D-GalA standards (Sigma-Aldrich,). Monosaccharide hydrolysis products and sugar standards were subjected to HAPEC using a Dionex ICS2500 chromatograph (Dionex, Sunnyvale, CA, USA) equipped with a CarboPac PA20 analytical column (3 mm × 150 mm). The column was eluted with 2 mM NaOH (0.45 mL/min), and monosaccharides were monitored using a pulsed amperometric detector (Dionex).[17]

G. Macrophage Activation by Polysaccharide Fractions

Macrophage activation was evaluated by NO production using the Griess method.[18] Mouse macrophages (RAW264.7), purchased from the American Type Culture Collection (ATCC, Manassas, VA, USA), were cultured in Dulbecco's Modified Eagle's medium (DMEM) containing 100 U/mL penicillin, 100 μg/mL streptomycin, and 10% fetal bovine serum at 37°C in a 5% CO_2 humidified atmosphere. Aliquots (180 μL) of a RAW264.7 cell suspension (5×10^5 cells/mL) were dispensed into each well of a 96-well plate, together with 20 μL of the different test agents at 50, 200, and 500 μg/mL concentrations and incubated for 48 hours. PBS and lipopolysaccharide (1 μg/mL) served as negative and positive controls, respectively. NO production was determined by measuring nitrite (NO_2^-) levels in cell supernatants using a colorimetric assay based on the Griess reaction. Supernatants (100 μL) were reacted with 50 μL Griess reagent at room temperature for 10 minutes, and nitrite was determined by measuring the absorbance at 540 nm using $NaNO_2$ as the standard.

III. RESULTS AND DISCUSSION

A. Effect of Developmental Stage on Fruit Body Total Polysaccharide and Protein Levels

Polysaccharide levels in hot water extracts of *H. erinaceus* fruit bodies harvested at different developmental stages were consistently higher than the corresponding protein content (Fig. 3). Significant differences in the amounts of total polysaccharide and protein were recorded over the 6 defined developmental stages, with polysaccharide levels in-

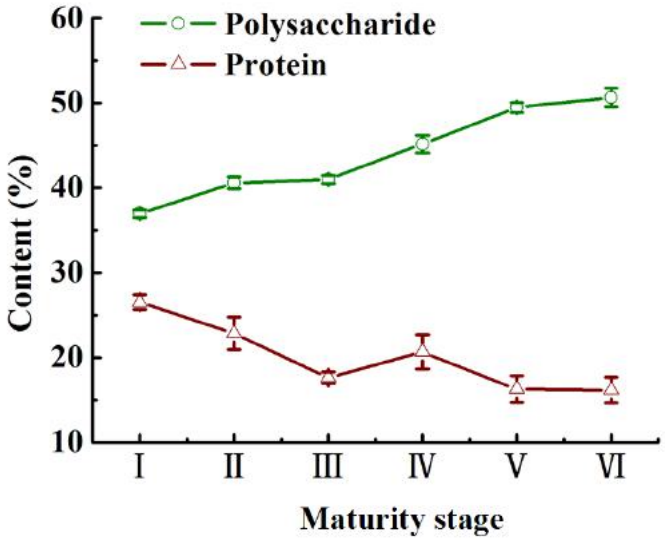

FIG. 3: Total polysaccharides and protein content of *Hericium erinaceus* fruiting bodies in different stages of maturity: stage I (bud-forming stage), stage II (small lump stage), stage III (split stage), stage IV (small fungal spine stage), stage V (mid-fungal spine stage), stage VI (mature).

creasing from 36.99% to 50.67%, whereas protein levels decreased from 26.56% to 16.18% (Fig. 3). These data contrast with those reported for *Lactarius* sp., in which polysaccharide levels decreased from 7.32% to 2.96%, and protein levels increased from 1.29% to 4.41% as the fruit bodies matured.[15] The proteins probably are transformed into amino acids by protein metabolism, and then amino acids are translated into polysaccharides by amino acid metabolism, resulting in the lowering of proteins' contents and the increasing of polysaccharides' contents in the most advanced stages. However, further research is needed.

B. Effect of Maturation Stage on Polysaccharide Yields

Polysaccharide levels and yields in the 9 fractions obtained from *H. erinaceus* fruit bodies harvested at stages IV, V, and VI are shown in Table 1. The amounts of polysaccharide precipitated with 50% EtOH increased, whereas those obtained with 30% EtOH decreased, with increasing sporophore maturation stage. Thus, extracts HP4A and HP6A,

TABLE 1: Polysaccharide Content and Yields in Extracts of *Hericium erinaceus* Fruit Bodies Harvested at Different Maturation Stages

EtOH concentration (%)	Stage	Fraction	Polysaccharide Content (%)	Yield (%)
30	IV	HP4A	49.00 ± 0.51	1.11 ± 0.06
	V	HP5A	31.21 ± 0.59	0.16 ± 0.05
	VI	HP6A	30.09 ± 0.54	0.18 ± 0.05
50	IV	HP4B	46.24 ± 0.81	1.08 ± 0.01
	V	HP5B	50.76 ± 0.52	1.09 ± 0.06
	VI	HP6B	67.14 ± 0.60	1.12 ± 0.04
70	IV	HP4C	45.58 ± 0.82	2.88 ± 0.01
	V	HP5C	42.30 ± 0.52	3.66 ± 0.06
	VI	HP6C	47.75 ± 0.89	3.42 ± 0.05

Values shown are the means ± standard deviations of 6 replicates.

obtained with 30% EtOH precipitation, contained 49.00% and 30.09% polysaccharide, respectively. Conversely, fractions HP4B and HP6B, precipitated with 50% EtOH, contained 46.24% and 67.14% polysaccharide, respectively. However, no consistent increasing or decreasing trend was evident with fractions obtained with the 70% EtOH precipitation.

C. Effect of Maturation Stage on Polysaccharide Molecular Weight Distribution Patterns

Polysaccharide molecular weight distribution patterns for the 9 fractions are shown in Fig. 4. HPLC of fractions HP4A, HP5A, and HP6A, obtained by 30% EtOH precipitation, revealed 3 separate peaks (a, b, and c) in each case (Fig. 4A). The molecular weights and percentages of the polysaccharides constituting each peak are summarized in Table 2. From Table 2, the maturation stage provided the differences of the molecular weights and percentages of every peak. For example, the average molecular weight of peak a in the polysaccharide fraction HP4A was 6.68×10^7 Da, and the area occupied 16.25% of total sample amount, whereas the average molecular weight of the same peak a in HP6A was 8.50×10^7 Da, and the area occupied 0.48% of total sample amount, respectively.

Fractions obtained by 50% and 70% EtOH precipitation all exhibited single symmetrical peaks with similar retention times (Fig. 4B, C) indicating the corresponding fractions were homogeneous polysaccharides. Peaks in the fractions obtained with 50% EtOH indicated a molecular weight of approximately 1.4×10^4 Da, whereas peaks in the 70% EtOH fractions indicated a slightly lower molecular weight of approximately 5.1×10^3 Da.

D. Effect of Maturation Stage on Monosaccharide Composition of Hydrolysed Polysaccharide Fractions

Monosaccharide components and their molecular ratios recorded after hydrolysis of the 9 polysaccharide fractions are listed in Table 3. All 9 fractions consisted of heteropolysaccharides containing fucose, galactose, glucose, and mannose. Galacturonic and gluconic acids were not detected in fractions HP4A, HP5A, and HP6A, indicating that these were composed of neutral polysaccharides, but were present in the other 6 fractions at concentrations ranging from 17.73 to 83.94 µmol/g (Table 3).

Monosaccharide compositions varied significantly in response to the maturation stage. Polysaccharides of stage VI fruit bodies contained the highest monosaccharide contents compared with those obtained by using same precipitation method from the fruiting bodies in other stages. For example, HP6B had the highest contents of fructose (365.60 µmol/g), galactose (1118.84 µmol/g), glucose (656.36 µmol/g), galacturonic acid (76.29 µmol/g), and gluconic acid (83.94 µmol/g). This is contrary to the results reported by Barros et al.,[15] in which the maturation stage did not significantly affect the monosaccharide compositions of *Lactarius* sp. mushrooms.

E. Effect of Maturation Stage on Nitric Oxide Production by RAW264.7 Macrophages

The antitumor effects of mushroom-derived polysaccharides involve various host immune responses, including the activation of the complementary system and macrophage-dependent systems.[19] Activated macrophages also play an important role in antitumor activity by secreting secondary compounds, such as NO and tumor necrosis factor-alpha (TNF-α), and by regulating the immune system to process and present antigens.[20]

All the polysaccharides examined in this study stimulated NO production by RAW264.7 macrophages (Fig. 5). Increased NO release and cytokine production by macrophages was also reported after exposure of the cells to a polysaccharide (HEB-AP Fr I) from *H. erinaceus* culture broths.[21]

In our study, fraction HP6A was most effective in stimulating dose-dependent NO production. Fractions derived from stage V and VI fruit bodies exhibited higher NO stimulatory activity,

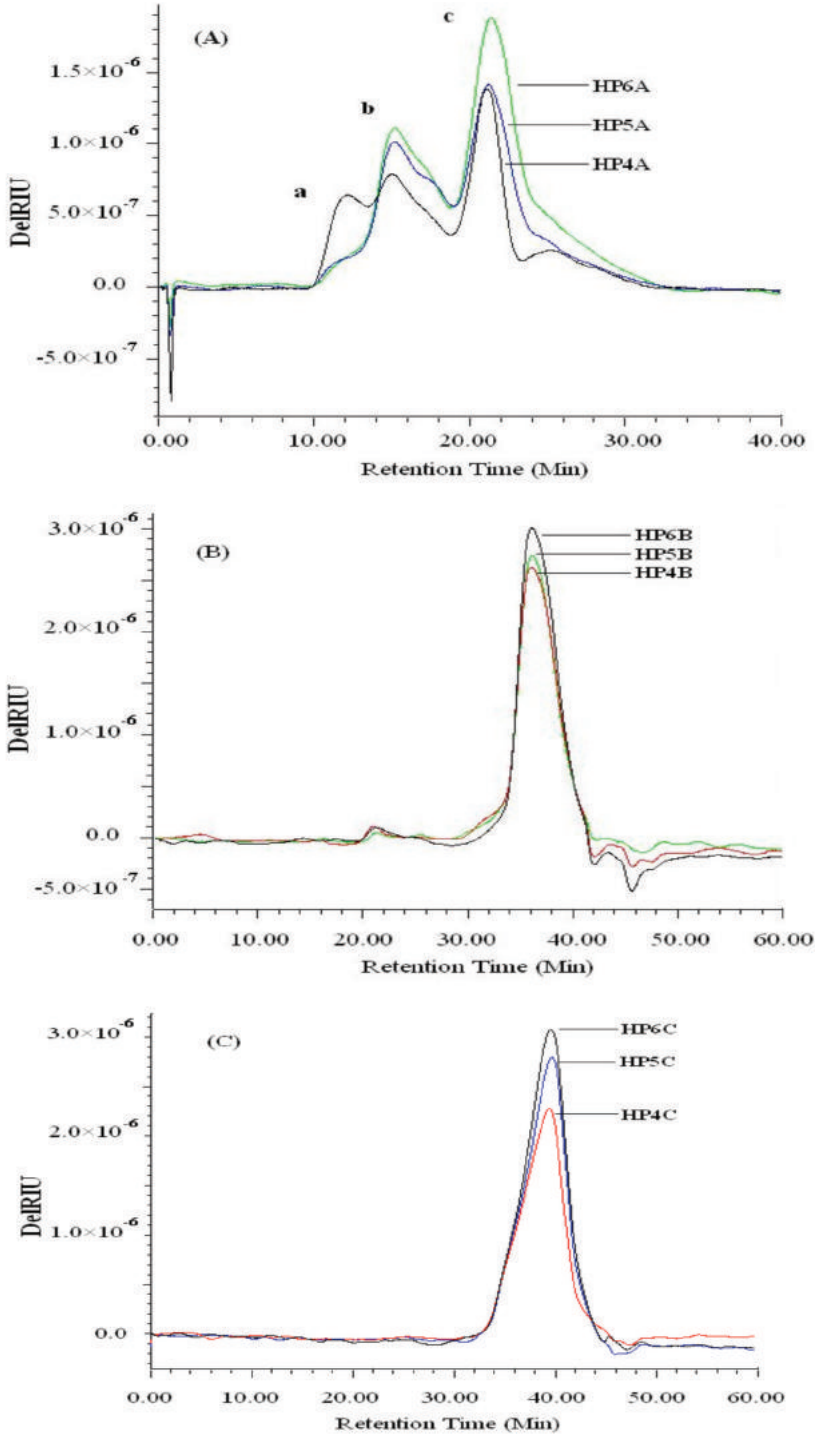

FIG. 4: Molecular weight distribution of 9 fractions from *Hericium erinaceus* fruiting bodies in (**A**) stage IV (small fungal spine stage), (**B**) stage V (mid-fungal spine stage), (**C**) stage VI (mature).

TABLE 2: Molecular Weight Distribution Patterns Among Fractions HP4, HP5, and HP6

	A							
Fraction	Peak a Mw (Da)	Ratio (%)	Peak b Mw (Da)	Ratio (%)	Peak c Mw (Da)	Ratio (%)	B *Mw* (Da)	C *Mw* (Da)
HP4	6.68×10^7	16.25	2.85×10^6	20.70	7.53×10^3	63.04	1.48×10^4	5.38×10^3
HP5	1.16×10^8	1.57	2.74×10^6	42.95	7.38×10^3	55.48	1.43×10^4	4.99×10^3
HP6	8.50×10^7	0.48	2.54×10^6	34.62	5.81×10^3	64.90	1.43×10^4	5.11×10^3

TABLE 3: Monosaccharide Components of Polysaccharide Fractions from Stage IV, V, and VI Fruit Bodies

	Monosaccharide					
Fraction	Fuc	Gal	Glc	Man	GalA	GluA
HP4A	58.49	207.34	481.06	68.33	—	—
HP4B	167.30	652.68	461.01	61.50	39.43	43.34
HP4C	64.77	182.68	465.10	114.45	45.9	47.63
HP5A	132.84	409.94	560.23	138.90	—	—
HP5B	174.60	689.11	383.69	57.22	31.73	37.81
HP5C	68.46	167.20	464.07	118.10	43.85	48.05
HP6A	147.62	464.49	567.95	188.85	—	—
HP6B	365.60	1118.84	656.36	159.14	76.29	83.94
HP6C	44.13	113.87	446.69	89.96	17.73	37.15

Values are expressed as μmol/g polysaccharide; —, not detected.

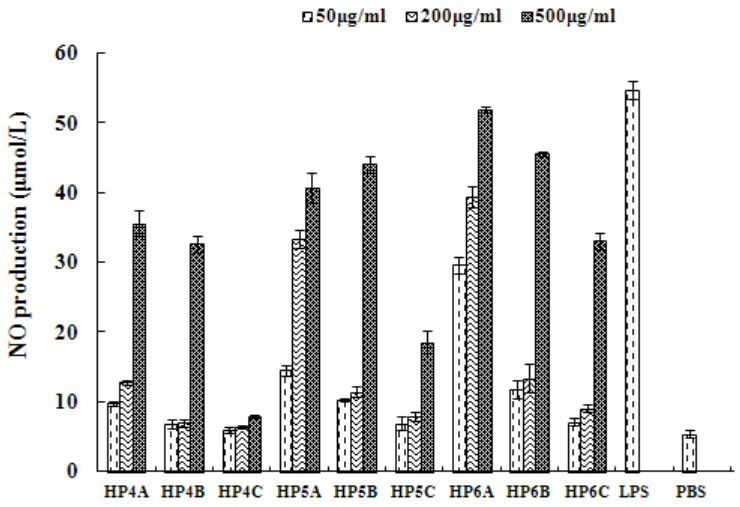

FIG. 5: Effect of fruiting body maturity stage on nitric oxide (NO) synthesis in murine macrophage-like cells RAW264.7. Cells (5×10^5 cells/mL) were stimulated by each polysaccharide fraction for 48 hours. Supernatants were collected and NO concentration was determined using the Griess reagent.

which may be linked to the higher polysaccharide levels recorded in these samples. Significant levels of macrophage activation was previously reported after exposure to extracts of 8 edible fungi containing high amounts of polysaccharide.[22]

Fractions obtained by 30% EtOH precipitation exhibited higher NO stimulating activity compared with the 50% and 70% EtOH fractions, which might be attributable to the higher molecular weight and neutral sugar composition of the former. A polysaccharide (PPS) from *Polyporus umbellatus*, with a high molecular weight of approximately 1.6×10^5 Da and consisting of the neutral sugar residues of glucose, galactose, and mannose also showed pronounced macrophage activation effects.[23]

IV. CONCLUSIONS

The knowledge about the changes of polysaccharide characteristics in different sporophore maturation stages will be a reference for finding the suitable stage to better develop functional and nutritional products. In the present study, the effect of maturation stages of the *H. erinaceus* fruiting body on the chemical compositions, polysaccharide molecular weight distribution, monosaccharide compositions, and macrophage activation activity were well investigated. Based on the results obtained, fractions derived from stage V and VI fruit bodies exhibited higher yields of polysaccharides and NO stimulatory activity. The molecular weight distribution and monosaccharide compositions varied significantly in response to the maturation stage of *H. erinaceus*, especially for fractions obtained by 30% EtOH precipitation. Detailed structure analysis of these 9 polysaccharide fractions is ongoing to address the correlation between structure and growth stage.

ACKNOWLEDGMENTS

This work was supported by funding from National Science & Technology Pillar Program during the Twelfth Five-Year Plan Period (2012BAD36B05), the Shanghai Academy of Agricultural Sciences Fund for Distinguished Young Scientists (2012–14), the Natural Science Foundation of China (31101269), the China Postdoctoral Science Special Foundation (2013T60648), and the China Postdoctoral Science Foundation (2012M511702). We thank Dr. John Buswell for linguistic revision of the manuscript.

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